

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<think>

- Zn content in Zn₂Fe(CN)₆ can be calculated by taking ratio of the weight of Zn to the weight of Zn₂Fe(CN)₆.
- Weight of Zn in Zn₂Fe(CN)₆ = 2 * 65.4 g = 130.8 g.
- Weight of Zn₂Fe(CN)₆ = 342.6 g.
- The ratio of Zn's weight to Zn₂Fe(CN)₆'s weight is (130.8 / 342.6) = 0.3815 approx.
- Therefore , Zn from 2 g rock = 0.969 g Zn₂Fe(CN)₆ * 0.3815 = 0.37 g Zn.
- The % of Zn in sample = (0.37 g Zn / 2 g rock) * 100 = 18.5 % ≈ 18% approx.

</think>

<answer>

F. 18 % Zn

</answer>